WeanAge user guide

Estimating weaning age from $\delta^{15}N$ in tooth dentine and bone collagen

Version 0.2 · October 2026 · https://github.com/andymaehartley/WeanAge

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About this guide

WeanAge estimates when young mammals were weaned from the nitrogen isotope ratio ($\delta^{15}N$) of their collagen. It works with two kinds of material: thin slices along one tooth, which record one animal’s childhood, and the bones of many young animals from one site that died at different ages. For cattle it also reads enamel $\delta^{18}O$, $\delta^{13}C$ and $^{87}Sr/^{86}Sr$ along a molar for the season of birth, the development of the rumen, the diet and moves. Further calculations put weaning in context: kill-off profiles from aged mandibles, the make-up of the diet from bone collagen, and, for people, the share of breast milk at each dentine increment, age at death from dental development, the timing of enamel hypoplasia, life tables and mobility. It runs as a local app or in a web browser.

This guide explains each part of the software, how the parts combine into a weaning age, and where each method comes from. Section 1 gives the background, sections 2–11 follow the calculations step by step, section 12 shows the app, and section 13 lists what is still provisional. Words in `code font` are column names, settings or files in the software. References are listed at the end.

1. Background

Why $\delta^{15}N$ records weaning

$\delta^{15}N$ rises by about 3–4‰ with each step up a food chain (DeNiro & Epstein 1981; Minagawa & Wada 1984). A suckling animal feeds on its mother’s tissues through her milk, so its own collagen is raised above hers, usually by 2–3‰ in practice (Balasse et al. 2001; Fuller et al. 2006). When milk is replaced by solid food, new collagen gradually falls to the adult level. The age at which that fall happens is the weaning age.

Collagen is the protein measured. Dentine collagen forms in layers as the tooth grows and is not renewed afterwards, so a series of slices from crown tip to root is a timeline of one animal’s diet (Beaumont et al. 2013). Bone collagen, by contrast, is renewed throughout life, so a bone records a mix of recent and older diet; a single bone cannot give a timeline, but many young animals that died at different ages can (Tsutaya & Yoneda 2013).

What “weaning age” means here

Weaning is a process, not an instant. WeanAge reports three milestones of the $\delta^{15}N$ fall (Figure 1): **onset** (5% of the fall has happened), **midpoint** (half of it) and **completion** (95%). The midpoint is reported by default: the curve is steepest there, so it is dated most precisely. Completion depends on how slowly $\delta^{15}N$ levels off and can fall well after suckling stopped. In known-age calves the $\delta^{15}N$ fall traces the end of suckling (Balasse et al. 2001), and the same has been found in sheep (Díez-Canseco & Tornero 2026).

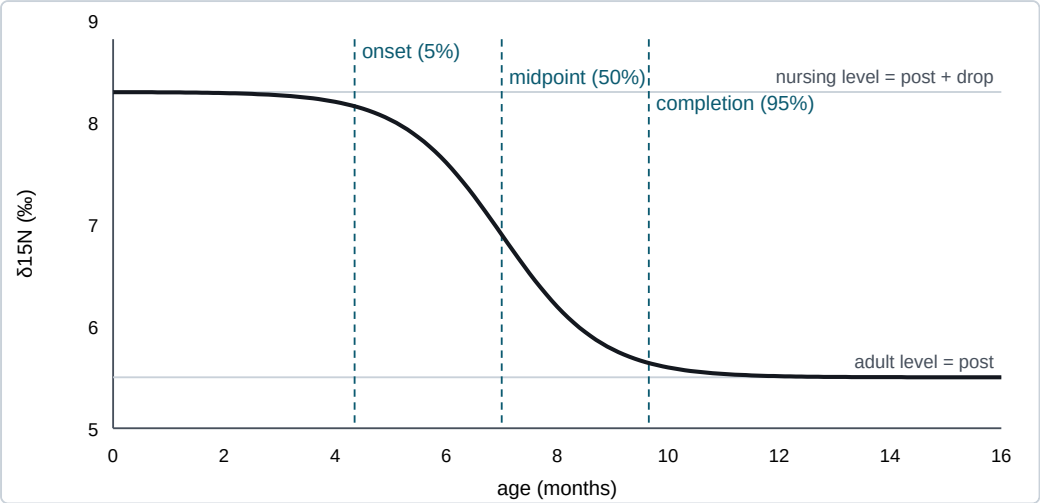


Figure 1. A logistic weaning curve and its three milestones, with the default 5% threshold.

The routes through the software

Tooth calculation
slices of one tooth
→ position to age (§3)
→ quality control (§2)
→ curve models (§4)
→ uncertainty, averaging (§4)
→ one animal’s weaning age

Bone calculation
many young animals + adult females
→ quality control (§2)
→ diet and bone-turnover model (§5)
→ Bayesian fit (§5)
→ a site’s weaning age

Enamel (cattle)
$\delta^{18}O$, $\delta^{13}C$, $^{87}Sr/^{86}Sr$ up one molar
→ yearly $\delta^{18}O$ cycle (§6)
→ season-of-birth index
→ rumen development, C4 share
→ moves, and months of the year (§6)

Context
aged mandibles → kill-off profile (§7)
bone collagen → diet make-up (§8)
people: dental age, hypoplasia, life tables, mobility (§9)

Then
compare groups (§10); plan sampling (§11)

The app has the same order: **Data → Settings → Results → Compare**, plus **Plan sampling**. On the Data tab a table is dropped first; WeanAge then lists the calculations it can feed, and any number of them can be ticked and run.

2. Data and quality control

Input. One CSV or Excel row per tooth slice, or per animal in bone mode; samples can also be typed in by hand. Column names are matched loosely (`Taxon` for `species`, `δ15N` for `d15N` and so on). Tooth data need `specimen_id`, `site`, `species`, `tooth`, `position_mm` (distance from the cusp apex, the tip of the crown, to the middle of the slice) and `d15N`. Bone data need `specimen_id`, `site`, `species` and `d15N`, plus an age at death (`age_months`, or `age_min_months` and `age_max_months`) for each young animal and `age_class = adult` and

`sex` for adults. Every row is checked as it is read (`ingest.py`, `schema.py`); rows that fail are listed with the reason, and the rest are kept.

Collagen quality control (`qc.py`). Collagen that has decayed or been contaminated gives misleading isotope values. A sample is flagged when its atomic C:N ratio, %C, %N or collagen yield is outside set limits, and flagged samples are plotted but left out of the fit. Two published sets of limits are offered:

preset	C:N	other limits	use for
Archaeological	2.9–3.6	%C ≥ 13, %N ≥ 4.8, yield ≥ 1%	archaeological collagen (DeNiro 1985; van Klinken 1999)
Modern mammals	3.00–3.28	as above	modern reference animals (Guiry & Szpak 2020)

If C:N is missing it is computed from %C and %N as (%C / %N) × (14.007 / 12.011).

3. Turning tooth position into age

A slice is measured as a distance along the tooth, but a weaning age needs an age. The **growth table** (`growth.py`, `data/growth_tables.json`) gives the age at which dentine formed at several points along a species' tooth; between the points the age is read off a straight line. Beyond the table, ages are extended along the last segment and the samples are marked as extrapolated.

tooth	ages covered	source
Cattle M1	about −2 to 2.5 months	crown completion from Brown et al. (1960); start age and height are placeholders
Cattle M2	about 1 to 12.5 months	crown start and end from Brown et al. (1960); crown height is a placeholder
Human M1	birth to about 9.5 years	Beaumont et al. (2013), after the London Atlas (AlQahtani et al. 2010); provisional
Pig M1, horse M1	—	placeholders

Because an M1 forms too early to show weaning in cattle, an M2 (or an M1 and M2 from the same animal, joined by `individual_id`) is normally needed. Positions measured from the base of the crown are converted using `crown_height_mm`.

Calibration (`calibrate.py`). Labs with known-age reference teeth can build their own table from (position, age) pairs with a straight-line, quadratic or isotonic (any increasing shape) fit. The scatter around the fit becomes the table's **age uncertainty**, which is carried into the weaning-age intervals (§4).

4. The tooth calculation

Curve models

Each specimen's $\delta^{15}N$ series is fitted with five curves (`models.py`). All parametric curves share an adult level `post` and a nursing rise `drop`, and differ only in the shape of the fall:

model	shape	milestones
Logistic	smooth S-shaped fall	from the centre and width, at 5/50/95% of the fall
Piecewise linear	flat, straight fall, flat ("broken stick")	the two break points and their midpoint
Exponential	flat, then fast fall that slows	start, start + τ·ln 2, start + τ·ln 20
Two-phase	two S-shaped steps with a plateau between	first step's onset, half the total fall, second step's completion
Elbow (WEAN-style)	4th-degree polynomial	the "elbow" after the $\delta^{15}N$ peak (completion only)

The logistic and piecewise curves are standard descriptions of a gradual and an abrupt transition. The exponential describes a diet change that is fast at first. The two-phase curve describes staged weaning: supplementary feeding while suckling continues, then final weaning, as expected where people took part of the milk (Balasse & Tresset 2002). The elbow model re-implements WEAN (Ganiatsou et al. 2023), so results can be compared with that tool; by default both axes are rescaled to 0–1 before the elbow is found, so the answer does not depend on units.

Curves are fitted by least squares with SciPy (`curve_fit`, bounded trust-region method) from 10–20 starting points, keeping the best fit. If the lab's analytical precision is given, it is used as the measurement error of each point.

When no age is reported

A curve can be fitted to any series, including a flat and noisy one, so WeanAge refuses to read a weaning age from it when:

- the fitted fall is smaller than the minimum (default 1‰): **no clear weaning signal**;
- the fall is centred before birth, which cannot be weaning;
- for the elbow model, $\delta^{15}N$ never falls after its peak, or the elbow's 95% interval covers most of the sampled ages.

The curve is still drawn so the data can be inspected. Falls larger than one food-chain step (default 5‰) are flagged, because they suggest another diet change or mixed-up samples.

Ages beyond the samples. A curve can be extended past the last slice, but its shape there is not measured. An age later than the last sample is shown only as **"after X months"** (or "before X months"), without an interval, and is left out of group comparisons.

Uncertainty

Each milestone has a 95% interval. By default, 1000 sets of curve parameters are drawn from the fit's covariance and the milestones recomputed for each; a residual bootstrap (Efron & Tibshirani 1993) can be chosen instead. The growth table's age uncertainty is added as one shared random shift per draw, because an error in the age model moves every slice of a tooth together. The interval is the middle 95% of the draws.

Choosing and averaging models

Models are compared by AICc, Akaike's information criterion corrected for small samples (Hurvich & Tsai 1989). The best-supported model has the lowest AICc. **Akaike weights** turn the AICc differences into shares of support that add up to 1, and the **model average** weights each model's age by its share (Burnham & Anderson 2002). Its interval mixes the models' random draws in the same proportions. An average is given only when the models that report an age hold at least half of the support. The headline result is the best model's age, with the model average beside it (Figure 2).

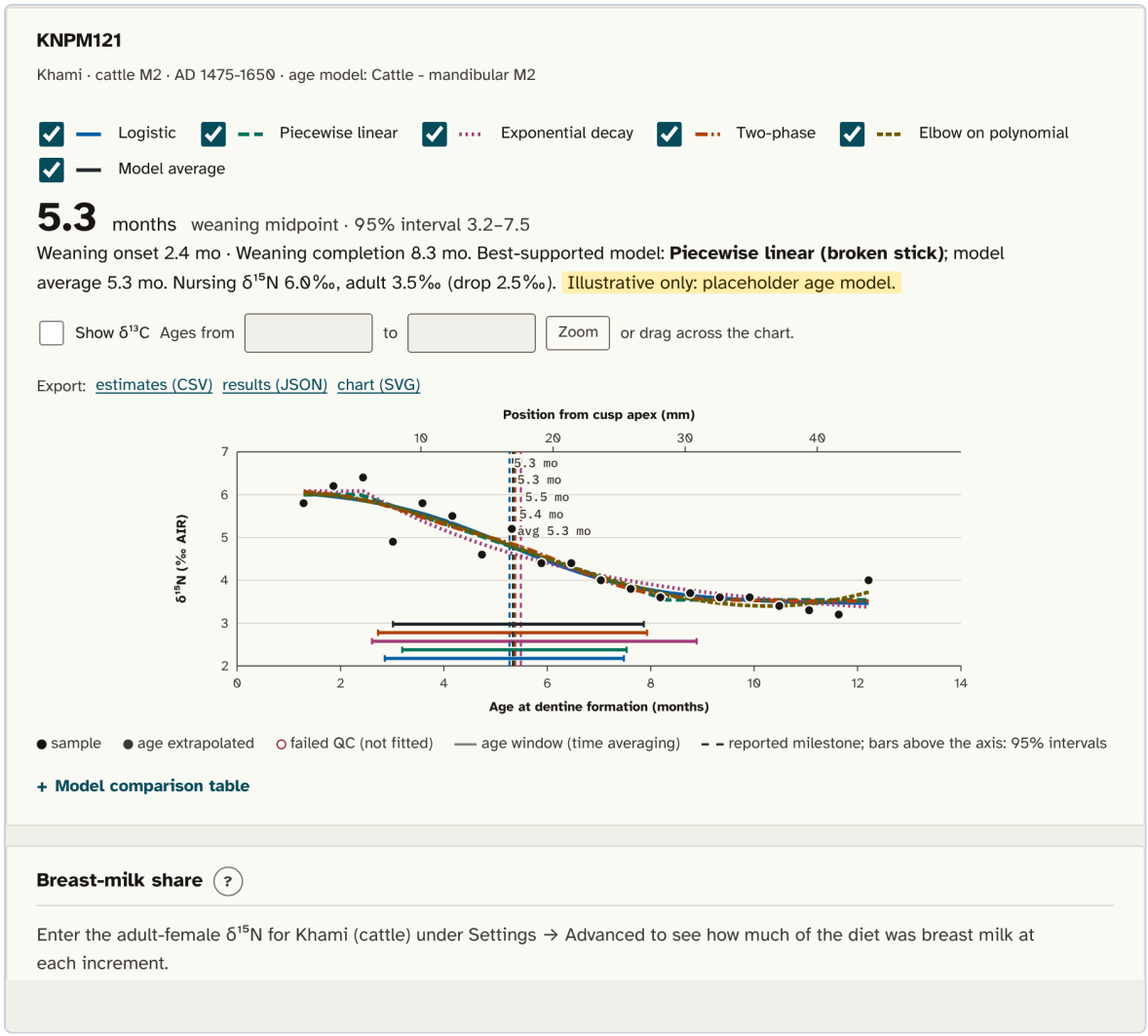


Figure 2. Tooth results for a Khami cattle M2 (real data from IsoArcH; slice positions assumed). The five model curves, their midpoints (dashed lines) and 95% intervals (bars above the age axis).

Corrections that can be switched on

- **Time averaging** (after Tsutaya 2020). A slice mixes dentine formed over several weeks: it has a height, and growth layers run at an angle through its depth. WeanAge covers each slice with an 8 × 8 grid over height and depth, converts each point to an age, and fits the sample as the weighted average of the curve over the grid, with less weight towards the pulp, where there is less dentine. The fitted curve then describes the underlying diet change, which is sharper than the blurred measurements.
- **Seasonal term.** A yearly wave, a·sin(2πt/12) + b·cos(2πt/12), is added to every parametric curve, so a pasture or fodder cycle is not mistaken for weaning. Prehistoric cattle calved seasonally (Balasse et al. 2021), so diet can change with the seasons in the first year.
- **Adult-female baseline** (after WARN; Tsutaya & Yoneda 2013). If the $\delta^{15}N$ of adult females from the same site is entered, results also give the nursing enrichment above them, and the adult plateau can be held within ±2 SD of the female mean.

Breast-milk share

When the adult females' $\delta^{15}N$ (and $\delta^{13}C$) is known for a site, each increment is read as a mix of collagen made on breast milk alone, one nursing enrichment above the adult females ($\delta^{15}N$; about 1‰ in $\delta^{13}C$, Fuller et al. 2006), and on the adult diet alone, as WARN does for bone (Tsutaya & Yoneda 2013). The milk share at each increment and its 95% interval are computed exactly on a grid of all mixes, allowing for the uncertainty of the baseline and the enrichment, and the age at which milk falls below half the diet is reported. A third weaning food (e.g. animal milk) can be added when $\delta^{13}C$ is measured.

Warnings

Warnings appear with each result. Besides those above, WeanAge flags:

- a profile that rises with age (Spearman ρ > 0.3, p < 0.05), since stress and starvation can raise $\delta^{15}N$ (Beaumont & Montgomery 2016);
- for cattle, $\delta^{13}C$ falling by 0.5‰ or more across weaning, because in known-age calves $\delta^{13}C$ rose as $\delta^{15}N$ fell (Balasse et al. 2001), so a fall suggests a different diet change;
- a yearly cycle in the curve's residuals or in $\delta^{13}C$ (an F-test for a sine wave, p < 0.05 and at least 0.2‰ or 0.3‰), which may be seasonal diet rather than weaning;
- **stress episodes**: two or more neighbouring slices whose $\delta^{15}N$ is above the fitted weaning curve (by 0.5‰) while $\delta^{13}C$ is below its expected level (by 0.3‰). Starvation and illness raise $\delta^{15}N$ without raising $\delta^{13}C$, whereas a change of food usually moves both the same way (Beaumont & Montgomery 2016). The expected $\delta^{13}C$ allows for a trend and a step at weaning; the correlation of the two leftovers is also reported;
- **$\delta^{34}S$ shifts**, when a `d34S` column is given: a change of level along the tooth, chosen by BIC among no, one or two shifts and reported if at least 2‰. $\delta^{34}S$ follows local geology and sea spray, so a shift suggests a move or a change of diet (Nehlich 2015).

5. The bone calculation

The bone calculation follows WARN (Tsutaya & Yoneda 2013), which was built for human skeletons, and adapts it for animals (`bone.py`). Each site × species (× period) group is fitted on its own. It combines three parts.

Diet. Collagen made at age *t* has $\delta^{15}N = (1 - p)(F + E) + p \cdot W$, after Millard (2000). F is the mothers' mean (adult females), E the nursing enrichment, W the $\delta^{15}N$ of collagen made only from weaning food, and p the share of non-milk food. p is 0 before weaning starts (t1), 1 after it ends (t2), and rises between them; WARN's default shape, $((t - t1)/(t2 - t1))^2$, is used, and linear or reverse shapes can be chosen.

Bone turnover. At birth, bone equals the mothers' value plus a newborn offset b. Every quarter of a month a share of the bone is rebuilt from the current diet, so bone $\delta^{15}N$ follows the diet with a delay. The shares come from turnover tables (`data/bone_turnover.json`). For humans these are WARN's rib collagen values (its Table 1), read as the share replaced per year and converted to monthly shares; this reproduces WARN's 96.6% replaced by 31 weeks. No turnover rates are published for livestock; the cattle, sheep, goat and pig tables are placeholders anchored to young fur seals, whose bone collagen is fully renewed in about 8–10 months (Newsome et al. 2006).

Fit. t1, t2, E, W, F, b and the scatter between animals are estimated together, and each animal's age at death is allowed to vary within its range. WARN fits with approximate Bayesian computation (Sisson et al. 2007); WeanAge writes down the likelihood and samples it with a Markov chain Monte Carlo sampler (Metropolis steps plus differential-evolution moves between 12 chains; ter Braak 2006), checking convergence with $\hat{R}$ (Vehtari et al. 2021). Priors follow WARN where it applies (W ~ N(F, 3), scatter ~ |N(0, 1)|; human E ~ N(1.9, 0.9) from Waters-Rist & Katzenberg 2010). For livestock, E ~ N(2.5–3.0, 1.0) and b ~ N(1.0, 1.0) are placeholders drawn from animal studies (Jenkins et al. 2001; Díez-Canseco & Tornero 2026); t1 and t2 have flat priors up to a user-set latest end of weaning, to avoid importing human weaning ages. t2 is reported as the end of suckling (§1).

Checks. Reynard & Tuross (2015) found that a raised $\delta^{15}N$ in the very young is common but weaning ages are hard to extract, often because samples are small. Before fitting, the youngest third of the young animals are compared with the mothers (one-sided Welch t-test); if they are not clearly higher, a warning says the data may hold no nursing signal. Warnings also appear for fewer than 10 young animals, intervals that only repeat the prior, and $\hat{R}$ above 1.1. As Tsutaya & Yoneda stress, the model always fits something, so look at the plot first.

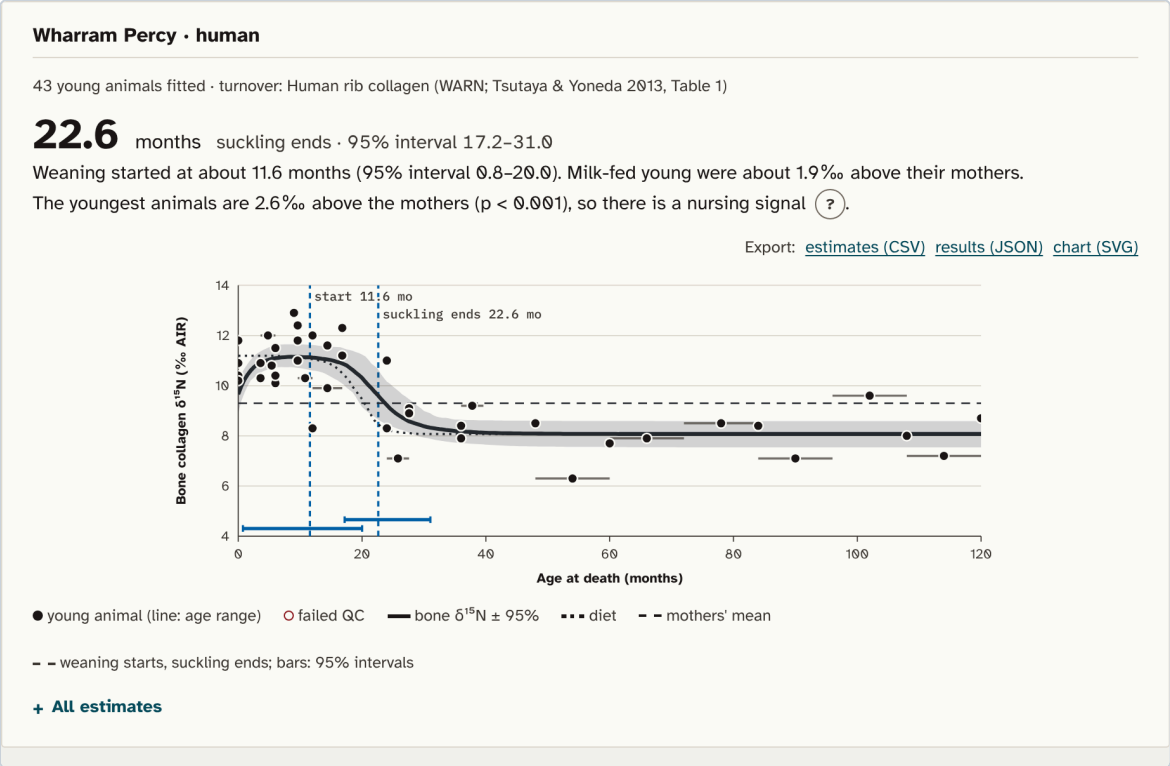


Figure 3. Bone results for medieval Wharram Percy (Richards et al. 2002, via CIMA). Points are children's rib collagen with their age ranges; the solid line is the modelled bone $\delta^{15}N$, the dotted line the diet. Suckling ends at 22.6 months (95% interval 17–31), matching the authors' estimate of about two years.

6. Season of birth, rumen development and C3/C4 diet (enamel)

This section uses tooth enamel rather than collagen, and is set up for cattle (`enamel.py`). Enamel of a molar forms from the cusp tip down to the enamel-root junction (ERJ) over several months, so samples drilled up the crown are a time series. Each sample's position is its distance from the ERJ (`erj_distance_mm`); values are enamel carbonate $\delta^{18}O$ and $\delta^{13}C$ (‰ VPDB).

Season of birth ($\delta^{18}O$). Enamel $\delta^{18}O$ follows drinking water, which is higher in summer. Each tooth's $\delta^{18}O$ series is fitted with a cosine, $\delta^{18}O(x) = M + A \cdot \cos(2\pi(x - x_0)/X)$, after Balasse et al. (2012): X is the crown length formed in a year and x_0 the position of the $\delta^{18}O$ maximum. The **birth index** x_0/X (0–1) places that summer on the tooth. Teeth of the same type start forming at about the same age, so differences in x_0/X between animals reflect differences in season of birth (cattle: Towers et al. 2014; Balasse et al. 2021). It is a relative index, not a date. For each site, the spread of the indices × 12 approximates how many months calving was spread over. No index is given when the $\delta^{18}O$ cycle is small (default under 0.5‰) or a cosine does not fit clearly better than a flat line.

Rumen development ($\delta^{13}C$). A suckling calf's rumen is not yet working. As it develops, fermentation enriches the carbon that reaches the enamel in ^{13}C (Passey et al. 2005; Cerling & Harris 1999), so first molar enamel $\delta^{13}C$ rises (Towers et al. 2014). Because plant $\delta^{13}C$ also changes with the seasons, $\delta^{13}C$ is fitted twice: with a yearly wave of the period found from $\delta^{18}O$, and with the same wave plus a rising step. The step is reported as rumen development only when it fits clearly better (AICc) and rises by at least 0.5‰; its position on the crown has a 95% interval. Rises above 3‰ are flagged, since rumen development alone gives about 1–2‰ and a change of diet (for example to C4 plants) may be involved.

C3 or C4 diet ($\delta^{13}C$). C4 plants (millet, maize, sorghum, many tropical grasses) are about 15‰ higher in $\delta^{13}C$ than C3 plants. Enamel is about 14.1‰ above the diet in large ruminants (Cerling & Harris 1999), so the diet's $\delta^{13}C$, from enamel formed after the rumen transition, is placed between a C3 and a C4 end member to give the share of C4 plants with a 95% interval. The default end members (C3 −27‰, C4 −11‰) are for a pre-industrial atmosphere; modern animals need modern values, about 1.5‰ lower, because fossil-fuel CO_2 has lowered plant

$\delta^{13}C$. C3 plants vary by several ‰ with water stress and shade (Kohn 2010), so the interval is wide; local plant or herbivore values narrow it.

Mobility ($^{87}Sr/^{86}Sr$). Strontium in enamel follows the rocks and soils where the animal fed (Bentley 2006). A change of level along the crown is found as for $\delta^{34}S$ above (BIC, at least 0.0005 by default); a step blurred over a few samples is reported once, as a gradual shift. Enamel averages strontium over months, so the real move was sharper, and may have been earlier, than the shift seen (Montgomery et al. 2010). Values outside a local range, if one is given, are flagged.

Months of the year. With the month of the $\delta^{18}O$ maximum (summer by default), any position can be placed in the year: enamel *d* mm below the maximum formed *d*/X of a year later. This dates the rumen transition and strontium shifts to a month, with an interval. Birth months need a reference tooth of the same type with a known birth month, because the birth index only orders animals: a birth one month later lowers the index by 1/12. Adding each animal's weaning age (from its dentine, or entered in `weaning_age_months`) then gives the month of weaning.

Smoothing. Enamel takes months to mineralise at any one position (Zazzo et al. 2005), so the signals are smoothed: the real $\delta^{18}O$ cycle was larger and the $\delta^{13}C$ change sharper than recorded. If the length of crown over which enamel matures is given, WeanAge treats each sample as a moving average over that length, divides the $\delta^{18}O$ amplitude by sinc(L/X), and narrows the rumen step accordingly. This follows the principle of Passey & Cerling (2002) in a much simplified form, so the corrected values are approximate. Positions are not converted into ages, because enamel formation tables for cattle are not yet available.

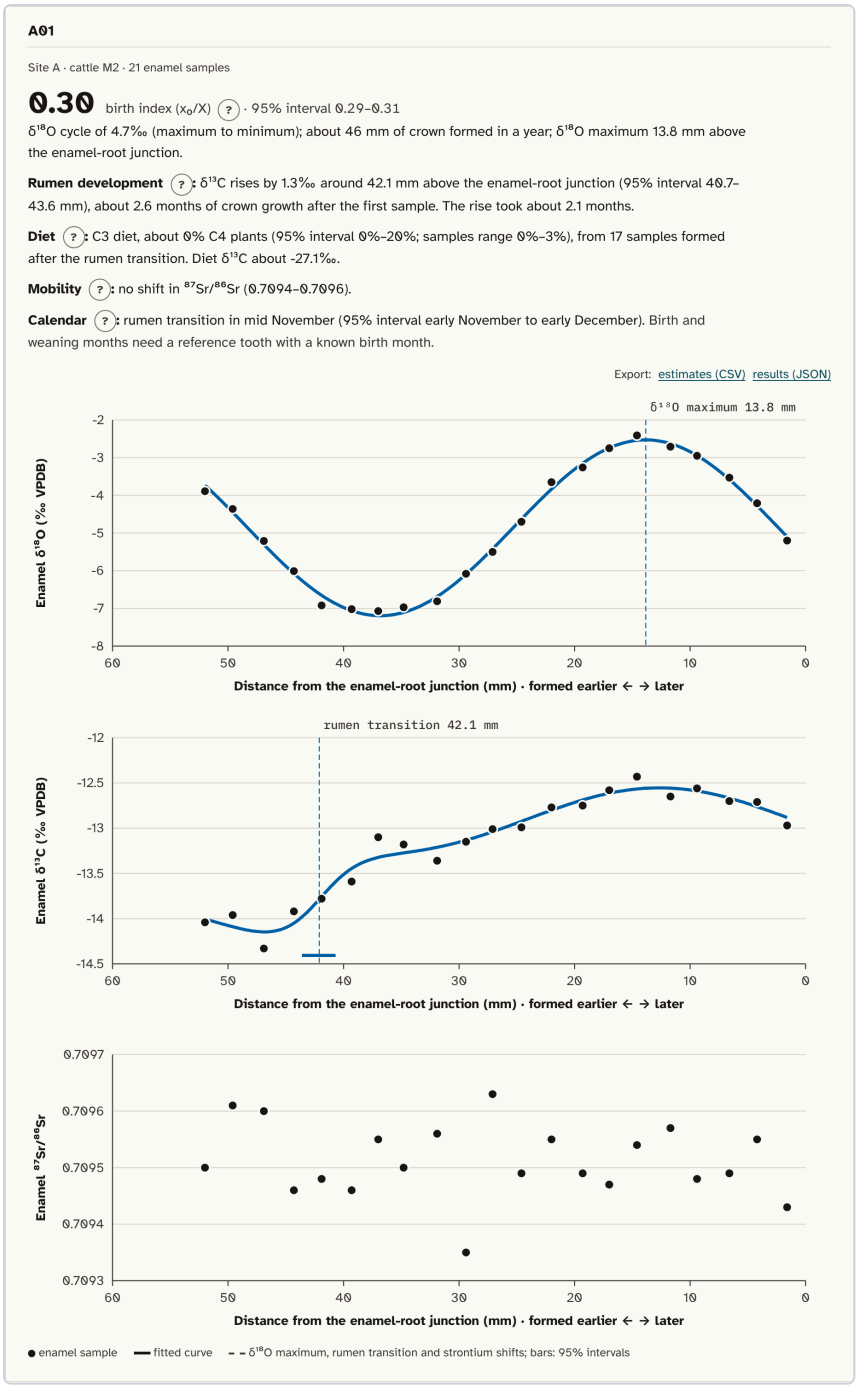


Figure 4. Enamel results for a synthetic cattle M2 (simulated data). From the top: $\delta^{18}O$ along the crown with the fitted cosine and its maximum; $\delta^{13}C$ with the fitted wave and rumen step, and the transition with its 95% interval; $^{87}Sr/^{86}Sr$, flat in this animal. The text above gives the share of C4 plants and the month of the rumen transition.

7. Kill-off profiles

At what ages a herd's animals were killed shows what they were kept for, and it is the backdrop to weaning: dairying, for example, may mean early weaning or culling of young (`killoff.py`). Each mandible is aged by tooth wear, as a Payne (1973) stage A–I for sheep and goats (A 0–2 months, B 2–6, C 6–12, D 1–2 years, E 2–3, F 3–4, G 4–6, H 6–8, I 8–10), or as an age range for any species. A mandible that could be in two stages ("D/E") is shared among them in proportion to the mandibles that fall in only one, as Payne did. For each site and species the profile gives the share killed in each class and the share still alive after it (survivorship), with 95% intervals from resampling the mandibles (Efron & Tibshirani 1993). Three summaries follow Payne's models of herds kept for milk (many deaths before 6 months), meat (6 months to 3 years) and wool, breeding or traction (many alive at 4 years); see Vigne & Helmer (2007) for how these models have been used and debated. A permutation test asks

whether two groups' profiles differ by more than chance. Young mandibles survive burial and excavation less well, so early deaths are often under-counted.

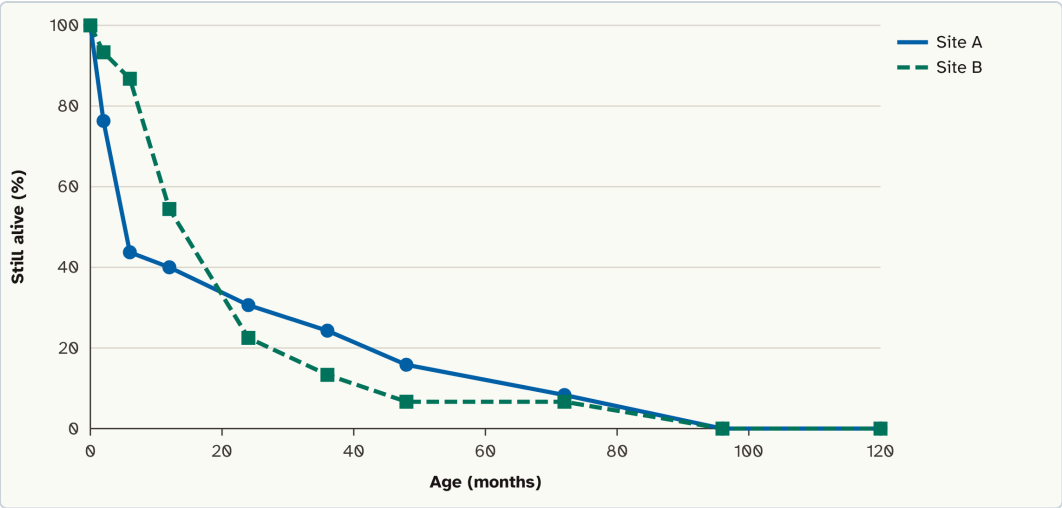


Figure 5. Survivorship of two synthetic sheep/goat herds (simulated data): Site A with most deaths in stages A–B, Site B in C–D.

8. Diet mixing

To say what weaned animals and people went on to eat, bone collagen $\delta^{13}C$, $\delta^{15}N$ and, if measured, $\delta^{34}S$ are split into the shares of a few food groups (`mixing.py`). Each isotope's expected value is $\Sigma\, p_s \mu_s + \Delta$, the share-weighted food values plus a collagen-diet offset, with a variance from the spread of each food, the offset and the measurement, as in FRUITS (Fernandes et al. 2014) and MixSIAR (Stock et al. 2018). Twenty thousand diets are drawn with every mix equally likely beforehand, and each is weighted by how well it explains the sample; the weighted draws give each food group's share with a 95% interval. Two warnings matter most (Phillips et al. 2014): with more food groups than isotopes + 1 the data cannot fix the shares, so the intervals stay wide; and a sample outside every mix of the food groups means a food is missing or a value is wrong. The bundled food values and offsets are rough placeholders. For $\delta^{15}N$, a controlled feeding study found collagen about 6‰ above the human diet (O'Connell et al. 2012); food values should come from plants and animals of the same site and period.

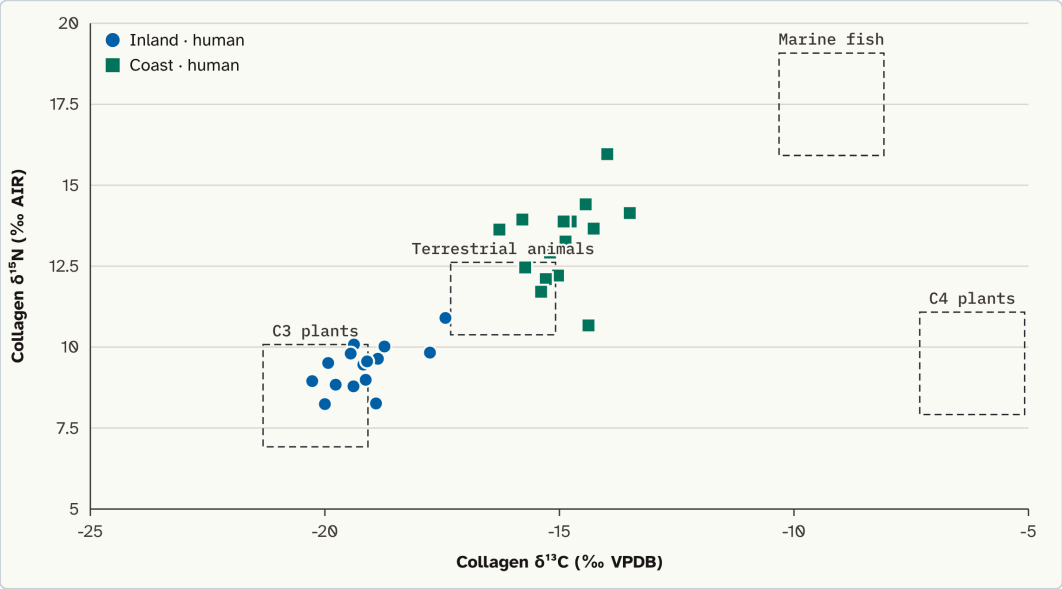


Figure 6. Synthetic human bone collagen from an inland and a coastal site (simulated data), with the placeholder food groups plus offsets (dashed boxes, ±1 SD).

9. People: dental age, hypoplasia, life tables and mobility

Dental age (`dental_age.py`). Children's teeth are scored in the formation stages of Moorrees et al. (1963), from crown initiation (Ci) to apex closed (Ac). A tooth in a stage lies between the ages at which it reached that stage and the next, so its age is the middle of that span. The teeth of a child are averaged, weighted by their precision; because they develop together, the combined uncertainty is that of a typical tooth. Teeth that disagree are flagged. The ages can be passed on as ages at death to the bone calculation and the life tables. The formation ages are placeholders until a published standard (AlQahtani et al. 2010; Smith 1991) is entered, so these ages are illustrative.

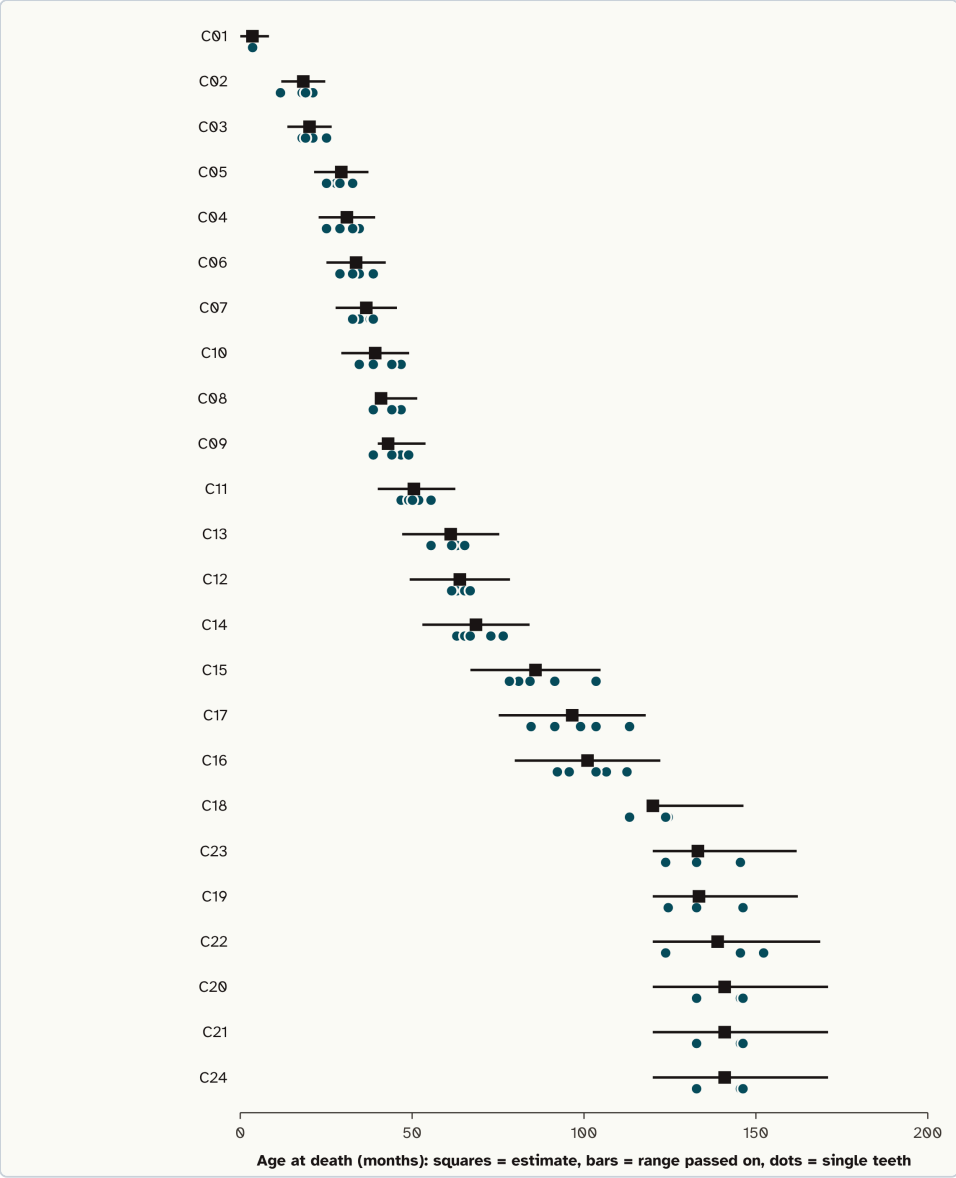


Figure 7. Dental ages of 24 simulated children: squares are estimates, bars the ranges passed on, dots the single teeth.

Enamel hypoplasia (`hypoplasia.py`). A defect's height above the cemento-enamel junction gives its age, taking the crown to form at an even rate from initiation to completion (Goodman & Rose 1990). Enamel forms more slowly towards the junction (Reid & Dean 2006), so these ages are approximate. Each defect is set against the same individual's weaning interval and stress episodes from the dentine, asking whether growth disruptions coincide with weaning.

Life tables (`lifetable.py`). Ages at death are shared over age classes to give deaths, survivors, the probability of dying and life expectancy in each class (Chamberlain 2006), with the share of infants and the juvenility index D5–14/D20+ (Bocquet-Appel & Masset 1977). A Siler (1979) hazard can be fitted directly to the age ranges. Single

years up to 5 show whether deaths cluster around weaning. A cemetery is not a living population: selective mortality and hidden frailty (the osteological paradox, Wood et al. 1992) and missing infant burials (Bocquet-Appel & Masset 1982) are flagged or stated with every result.

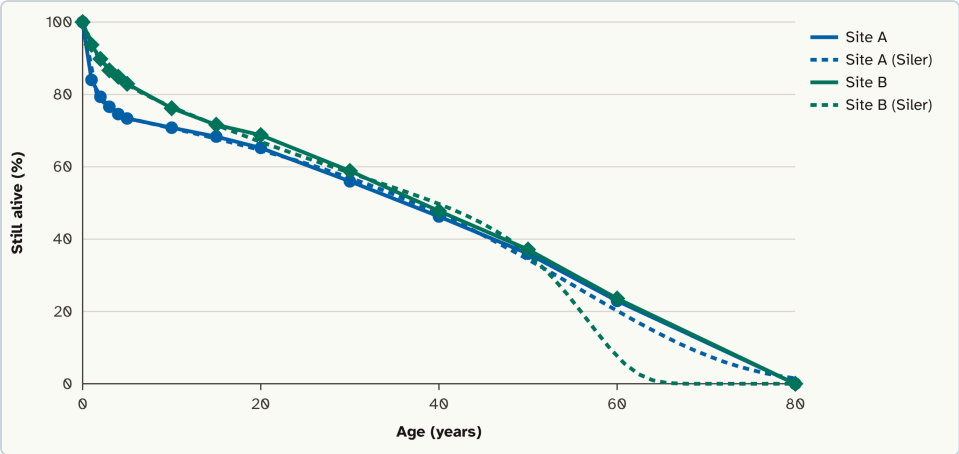


Figure 8. Survivorship in two simulated cemeteries, with Siler fits (dashed). Site B lost most of its infant burials.

Mobility (`mobility.py`). Bulk enamel $\delta^{18}O$ is converted to drinking water (carbonate: Chenery et al. 2012, after Coplen et al. 1983 for the scale; phosphate: Daux et al. 2008), with an SD of 1‰ for the conversion (Pollard et al. 2011). $^{87}Sr/^{86}Sr$ is compared with a local range from local fauna or plants (mean ± 2 SD; Price et al. 2002), or failing that with Tukey fences among the people (Lightfoot & O'Connell 2016). First molars and deciduous teeth may carry a breastfeeding signal in $\delta^{18}O$ (Wright & Schwarcz 1998).

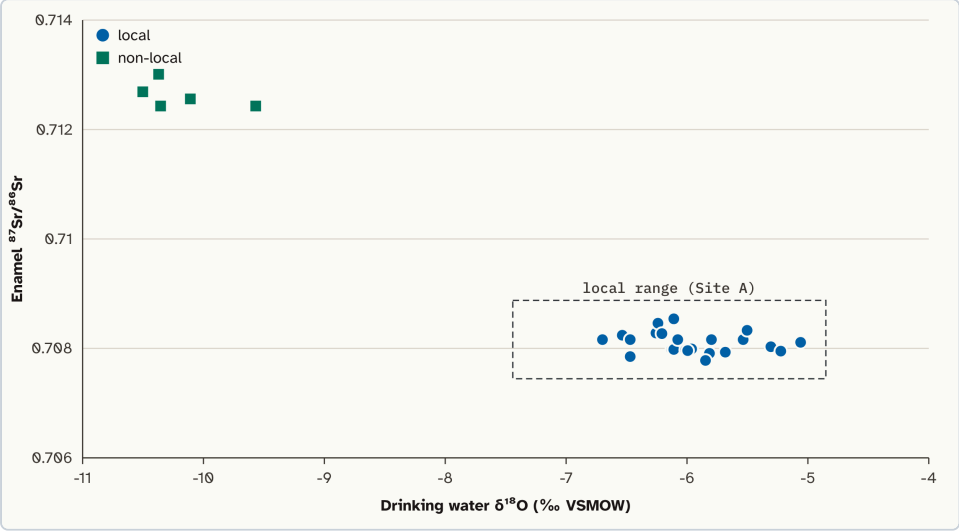


Figure 9. Drinking-water $\delta^{18}O$ and $^{87}Sr/^{86}Sr$ of 30 simulated people, with the local range; five grew up elsewhere.

10. Comparing groups

The Compare tab groups tooth estimates by site, species and/or period (`aggregate.py`, `inference.py`, `joint.py`).

- **Summary statistics**: n, mean, SD, median, range and a 95% confidence interval.
- **Random-effects means** (DerSimonian & Laird 1986): each group's mean age, allowing for real differences between animals (τ) as well as each estimate's own uncertainty, taken from its 95% interval. A Q-test asks whether groups differ at all; pairwise tests use Holm's (1979) correction for multiple comparisons; a permutation test (5000 shuffles of the group labels) checks the result without assuming normal errors.
- **Hierarchical Bayesian model** (two-stage): each specimen's estimate is treated as a noisy measurement of its true age, and true ages vary around a group mean. The between-animal SD has a half-Cauchy prior (Gelman

2006). It gives the probability that one group weaned earlier than another, and partially pooled specimen estimates.

- **One-stage model**: the logistic curves of all specimens and the group comparison are estimated together from the raw $\delta^{15}N$ values, so uncertainty in each curve's shape and each tooth's age model is carried through. It uses the logistic curve only.

Validation. If known weaning ages (`known_weaning_age_months`) or published estimates are included, the Compare tab reports each model's bias, mean absolute error, root-mean-square error, correlation and 95% interval coverage.

11. Planning a study

Both planners answer "would this sampling design work?" before teeth are cut or bones sampled.

Teeth (`simulate.py`). The planner invents teeth with known weaning curves, "cuts" them as the lab would (slice spacing and height, angled growth layers, measurement noise, lost samples, age-model error), runs the full analysis, and scores each model's completion age against the truth. Trying one setting at several values (for example 0.5, 1 and 2 mm slices) with the same random seed shows which protocol is accurate enough. Simulating curves of other shapes than the logistic shows how each model copes when its shape is wrong.

Bones (`bone_planner.py`). The planner invents sites with a known end of suckling and samples them as planned: the number of young animals, how their ages are spread, how precisely they are aged, and the number of adult females. It runs the bone calculation on each site and reports the error, the width and coverage of the 95% interval, and how often the nursing signal is found. Because the data are simulated with the model that is fitted, it tests the design, not the model; with placeholder turnover it shows relative, not absolute, precision.

12. Using the app

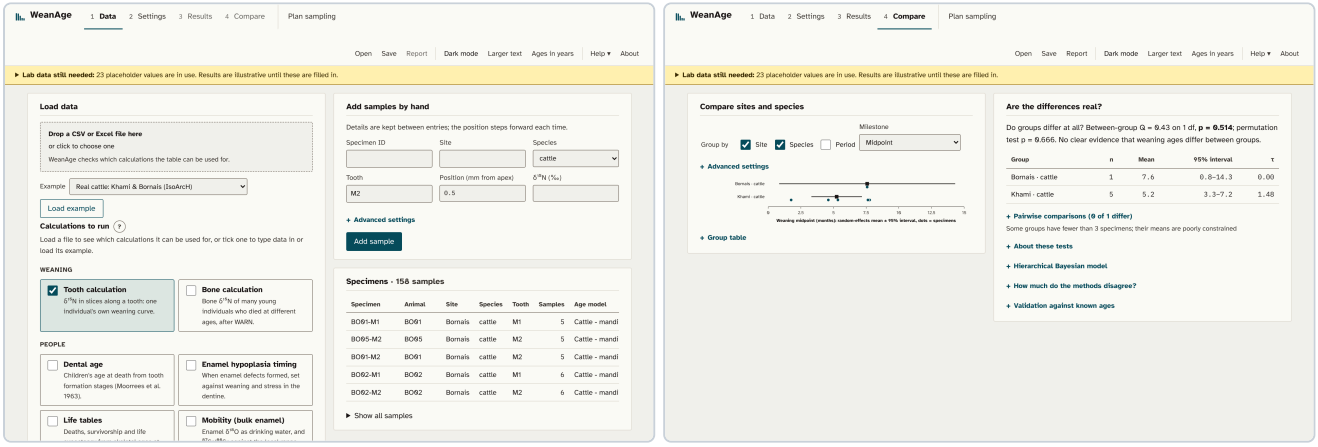


Figure 10. The Data tab with the cattle example loaded (left) and the Compare tab (right).

1. **Data.** Drop a file, then tick the calculations it can feed (or load an example, or type samples in). Problems are listed with their row numbers.
2. **Settings.** The main choices are shown; less used ones are under *Advanced settings*. The defaults suit a first run.
3. **Results.** Pick a specimen or group on the left. The headline gives the reported age and its 95% interval; the chart, the model table and the warnings explain it. Results can be exported as CSV, JSON or SVG, and the whole run as an HTML report.
4. **Compare** (tooth data) tests differences between groups.
5. **Plan sampling** runs the planners.

Save writes the data, settings and results to a project file; work is also kept in the browser. *Ages in years* switches every result to years. Every **?** opens a short explanation. Examples from IsoArcH (cattle teeth from Khami and Bornais) and CIMA (Cocozza et al. 2022; human bone from Wharram Percy, York Fishergate and Newark Bay, and human teeth from Mikulčice) are included; their slice positions or tooth age models are partly assumed, as noted when they load.

13. Limits and placeholders

Values that still need laboratory data are marked **placeholder** in the app and listed in the yellow banner at the top. Until they are replaced, results are illustrative. The most important are:

- crown heights and formation ages along the tooth for cattle, pig and horse, and a checked human M1 table;
- bone collagen turnover rates and nursing-enrichment priors for livestock;
- collagen quality-control limits and the minimum weaning drop for each laboratory;
- the food groups and collagen-diet offsets of the diet-mixing model;
- tooth formation ages for dental age and hypoplasia timing.

Other limits: growth layers are treated as straight and the tooth as round in cross-section; the one-stage model uses only the logistic curve; and the bone calculation is not yet part of the Compare tab or the report. Animals that died young are not a random sample of the herd (for example culled surplus calves in dairying), just as children who died are not a random sample of the living, which affects which ages are represented.

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Example data: IsoArcH (doi:10.48530/isoarch.2021.014, CC BY 4.0; Dyvart et al. 2018; Griffith & Mulville 2020) and CIMA (doi:10.48493/s9nf-1q80, CC BY-SA). WeanAge re-implements the methods of WARN, MDSS and WEAN from their publications; no code was copied.